The Pentium is faster than the 80286 system for four reasons, (1) the greater clock speed, (2) more lines in the data bus, (3) the addition of a memory cache, and (4) a more efficient internal design, requiring fewer clock cycles per instruction.

If the Pentium was a Cadillac, the TMS320C40 would be a Ferrari: less comfort, but blinding speed. This chip is representative of several microprocessors specifically designed to decrease the execution time of DSP algorithms. Others in this category are the Intel i860, AT&T DSP3210, Motorola DSP96002, and the Analog Devices ADSP-2171. These often go by the names: **DSP microprocessor**, **Digital Signal Processor**, and **RISC** (Reduced Instruction Set Computer). This last name reflects that the increased speed results from fewer assembly level instructions being made available to the programmer. In comparison, more traditional microprocessors, such as the Pentium, are called **CISC** (Complex Instruction Set Computer).

DSP microprocessors are used in two ways: as slave modules under the control of a more conventional computer, or as an imbedded processor in a dedicated application, such as a cellular telephone. Some models only handle fixed point numbers, while others can work with floating point. The internal architecture used to obtain the increased speed includes: (1) lots of very fast cache memory contained within the chip, (2) separate buses for the program and data, allowing the two to be accessed simultaneously (called a **Harvard Architecture**), (3) fast hardware for math calculations contained directly in the microprocessor, and (4) a *pipeline* design.

A **pipeline** architecture breaks the *hardware* required for a certain task into several successive stages. For example, the addition of two numbers may be done in three pipeline stages. The first stage of the pipeline does nothing but fetch the numbers to be added from memory. The only task of the second stage is to add the two numbers together. The third stage does nothing but store the result in memory. If each stage can complete its task in a single clock cycle, the entire procedure will take three clock cycles to execute. The key feature of the pipeline structure is that another task can be started before the previous task is completed. In this example, we could begin the addition of *another* two numbers as soon as the first stage is idle, at the end of the first clock cycle. For a large number of operations, the speed of the system will be quoted as one addition per clock cycle, even though the addition of any two numbers requires three clock cycles to complete. Pipelines are great for speed, but they can be difficult to program. The algorithm must allow a new calculation to begin, even though the results of previous calculations are unavailable (because they are still in the pipeline).

Chapters 28 and 29 discuss DSP microprocessors in much more detail. These are amazing devices; their high-power and low-cost will bring DSP to a wide range of consumer and scientific applications. This is one of the technologies that will shape the twenty-first century.